

Study on Port Business Intelligence System Combined with Business Performance Management

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Abstract—This paper describes the port BI system combined with BPM, which enables port enterprises to perform an in depth data analysis to support decision making and to promote performance management. Based on illustrations of the BPM strategies of port enterprises, the BI system architecture is proposed, and some key points in the knowledge discovery process of the BI system are discussed, which are all aligned with existing port BPM strategies.

Keywords—business intelligence (BI); business performance management (BPM); knowledge discovery

I. INTRODUCTION

Business intelligence (BI), which is a business management term used to describe applications and technologies that are used to gather, provide access to and analyze data and information about an enterprise, in order to help them make better informed business decisions, has been attracting a significant amount of research, industry, and media attention of late.^[1]

Business performance management (BPM) is a set of management and analytic processes, supported by technology, that enable businesses to define strategic goals and then measure and manage performance against those goals.^[2] Some commentators see BPM as the next generation of business intelligence.^[3] BI systems conduct enterprise data integration and data mining, which commit itself to technology service and decision-making support. Thus, BPM combined with BI system allows a systematic, integrated approach that links enterprise strategy to core processes and activities.

In this paper, we describe the port BI system combined with BPM, which enables port enterprises to perform an in depth historical analysis of the monitored metrics in order to detect any abnormal pattern or opportunities. We illustrate the business performance management strategies of port enterprises, and discuss some key points in the knowledge discovery process of the BI system, which are all aligned with existing port BPM strategies.

II. PORT BUSINESS INTELLIGENCE AND BUSINESS PERFORMANCE MANAGEMENT

A. Port Business Performance Management

In order to detect the changing business and customer needs, enterprises need the ability to monitor the business metrics and Key Performance Indicators (KPI) which are critical to its survival. Business Performance Management

(BPM) has emerged as critical solution to provide businesses the much needed technology to *sense* both business and IT metrics and *respond* to any threats and opportunities.^[4]

Presently, performance management has been adopted by many port enterprises to enhance the port enterprise competitiveness. An investigation on Chinese port BPM shows that 77.88 percents of the total investigated ports have already carried out the enterprise performance management.^[5]

B. Port BI System Combined with BPM

A critical component for improving business performance is the ability for the enterprise to take advantage of all available information, which becomes more difficult with the constantly increasing volume of information generated by transaction processing systems, such as ERP system. Hence, it provides a powerful combination to combine BI system with the BPM solution.

For port enterprises, the principle advantages of BI system combined with BPM can be summed up as follows:

- BI system can help control costs and provide an equitable basis for the development of pricing strategies, which help to improve the financial performance indicators of port enterprises;
- BI system will generate knowledge such as weather conditions or route occupancy status, so as to help reasonably developing port operation plan, which can decrease the waiting time of ships or other equipments, and help to improve the performance indicators such as customer satisfaction, port yard usage, and port throughput;
- BI system can provide a reasonable basis for setting indicators for the port Human Resources Performance Management.

III. THE PORT BUSINESS INTELLIGENCE SYSTEM ARCHITECTURE

Generally, a business intelligence system should have the following basic features:

- Data Management: including data extraction, data cleaning, data integration, as well as efficient storage and maintenance of large amounts of data.
- Data Analysis: including information queries, report generation, and data visualization functions.
- Knowledge Discovery: extracting useful information (knowledge) from the rapidly growing volumes of digital data in databases.

Since each industry requires different knowledge, there would be real significance only when the study of business intelligence is conducted for some specific industry environment.^[6] Thus, we propose the port BI system functional architecture based on the port business requirement, as described in Fig. 1.

IV. KNOWLEDGE DISCOVERY IN PORT BUSINESS INTELLIGENCE SYSTEM

With Business Intelligence system effectively implemented, executives can anticipate and respond to shifting market dynamics, intelligently allocate and utilize critical resources and consistently meet management and shareholder expectations.^[7] Knowledge discovery plays an important role in the BI system to improve the caliber and speed of decision making and to promote the business performance management. Thus, we propose the knowledge discovery process in port BI system from the viewpoint of BPM.

A. Building Data Repository

Port enterprises generally have a high level of information management. The implementation of ERP, CRM, and other systems has laid a good foundation for Business Intelligence System. Large amount of structured data in tabular form is stored in the database of the transaction processing systems, which can be easily obtained from the existing information systems.

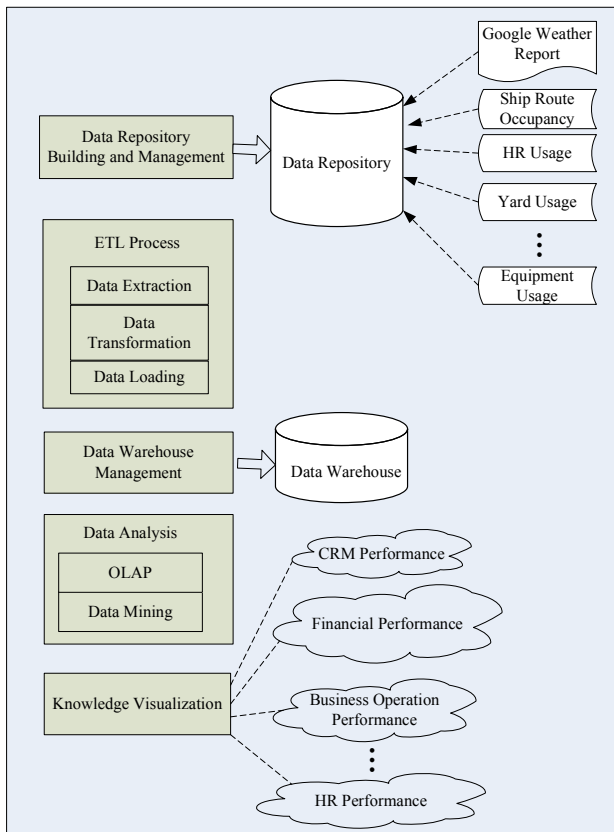


Figure 1. The Port Business Intelligence System Architecture

The port operation mainly involves service for ships entering the port and leaving the port, which has closely relationship with the port business performance. Accordingly, the structured data of port BI system includes all business data generated in the port service, such as charging terms in contract, dispatching note, operation records, freight tally note, bills, sea waybill, and so on.

On the other hand, in the category of business performance management, bringing together the worlds of structured and unstructured data can add significant value to the enterprise.^[7] For ports, unstructured data consists of two basic categories:

- Bitmap Objects: such as the ship route map, or the port yard map.
- Textual Objects: such as the google weather report, or the contract terms in text form.

Table I describes the contents of data repository in port Business Intelligence system.

B. Defining Data Warehouse Subject

General concern about performance management of port enterprises involves four perspectives:

- the customer perspective: including performance indicators such as market share, customer satisfaction, customer retention, the rate of customers
- business process perspective: including performance indicators such as the number of berths, yard area, mechanical equipment efficiency, the average working efficiency, the dispatching ability
- innovation and learning perspective: including performance indicators such as staff turnover rate, the rate of staff training, or equipment renewal rate
- financial perspective: including performance indicators such as the enterprise profitability, the business operation capacity, enterprise solvency and business development capacity.^[8]

We accordingly define the main data warehouse subjects in port BI system as customer subject, yard subject, equipment subject, and employee subject. Table II describes the data warehouse subjects in detail.

C. The Multidimensional Data Model

After defining the data warehouse subjects, we use multidimensional data model to describe the subjects.

The multidimensional data model is an integral part of On-Line Analytical Processing, or OLAP, which is designed to solve complex queries in real time.^[9]

TABLE I. DATA REPOSITORY CONTENTS OF PORT BI SYSTEM

Structured Data	Unstructured Data	
	Bitmap Objects	Textual Objects Subhead
• charging terms in contract • dispatching note • operation records • freight tally note • bills • sea waybill • ...	• ship route map • port yard map • ...	• weather report • contract terms in text form • ...

TABLE II. DATA WAREHOUSE SUBJECTS

Subject	Key	Attribute Group
Customer	Customer ID	Customer Type
		customer information
		contract fee
		arreage status
		...
yard	yard ID	Yard type
		Cargo type
		Storage amount
		...
equipment	equipment ID	Equipment type
		Work time
		Energy cost
		...
employee	employee ID	Work type
		salary
		Training status
		Work time
		...

The relational implementation of the multidimensional data model is typically a star schema, or a snowflake schema. A star schema is a convention for organizing the data into dimension tables, fact tables, and materialized views. Ultimately, all of the data is stored in columns, and metadata is required to identify the columns that function as multidimensional objects.

We choose port contract as main fact table in the customer subject, and design the star schema accordingly, as described in Fig. 2

V. CONCLUSIONS

This paper describes a port BI system combined with BPM, based on analysis on the BPM strategies of port enterprises. The system architecture is presented, and some key points in the knowledge discovery process of the BI system are discussed in particular. Further work include ETL process implementation, data warehouse physical design, knowledge visualization, and so on.

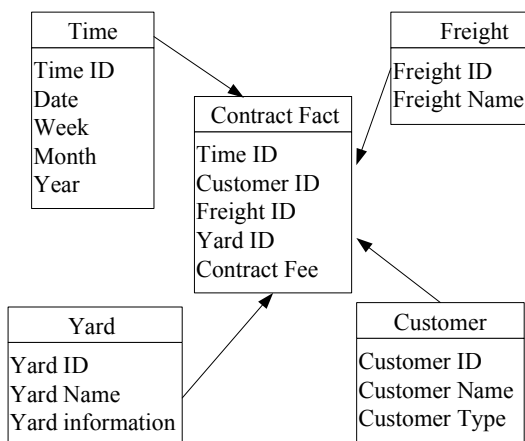


Figure 2. The Star Schema Diagram

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